Solution:

X	$\frac{1}{X}$
1	1.0000
0.5	2.0000
10	0.1000
45	0.0222
175	0.0057
0.01	100.0000
4.0	0.2500
11.2	0.0893
	$\Sigma\left(\frac{1}{X}\right)$=103.4672

Table : 8.7

$n = 8$

$$HM = \frac{n}{\Sigma\left(\frac{1}{X}\right)} = \frac{8}{103.467} = 0.077$$

Example 8.10

From the following data compute the value of Harmonic Mean

Marks	10	20	25	40	50
No. of students	20	30	50	15	5

Solution:

Calculation of Harmonic Mean

Marks X	No. of Students f	$\frac{f}{x}$
10	20	2.000
20	30	1.500
25	50	2.000
40	15	0.375
50	5	0.100
	$N = 120$	$\Sigma\left(\frac{f}{X}\right) = 5.975$

Table : 8.8

$$HM = \frac{N}{\Sigma\left(\frac{f}{X}\right)} = \frac{120}{5.975} = 20.08$$

Example 8.11

Calculate Harmonic Mean for the following data given below:

Value	0-10	10-20	20-30	30-40	40-50
Frequency	8	12	20	6	4

Solution:

Calculation of Harmonic Mean

Value	Mid-Point m	f	$\left(\frac{f}{m}\right)$
0-10	5	8	1.60
10-20	15	12	0.80
20-30	25	20	0.80
30-40	35	6	0.17
40-50	45	4	0.09
		N = 50	$\Sigma\left(\frac{f}{m}\right)$ **=3.46**

Table : 8.9

$$HM = \frac{N}{\Sigma\left(\frac{f}{m}\right)} = \frac{50}{3.46} = 14.45$$

Special applications of Harmonic Mean

The Harmonic Mean is restricted in its field of usefulness. It is useful for computing the average rate of increase of profits of a concern or average speed at which a journey has been performed or the average price at which an article has been sold. The rate usually indicates the relation between two different types of measuring units that can be expressed reciprocally.

For example, if a man walked 20km in 5 hours, the rate of his walking speed can be expressed

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